

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Assisted senior engineers with **site surveys** and **data collection** for industrial client energy assessments, gathering key performance indicators for energy consumption.
- Contributed to the analysis of energy consumption data using **Python scripting**, identifying inefficiencies in industrial utility systems and proposing initial energy saving opportunities.
- Supported the preparation of technical reports and engineering calculations for energy audits, aligning with standards such as **IS 399** and **EN 16247**.
- Gained hands-on exposure to core industrial utility systems including **HVAC**, **Steam**, and **Compressed Air**, documenting operational parameters during site visits.
- Collaborated with cross-functional teams to support **Measurement Verification (MV)** activities, tracking and validating project savings post-implementation.
- Contributed to internal research on energy efficiency best practices, benchmarking against industry standards like **ISO 50001** to improve technical processes.
- Containerized data analysis scripts using **Docker** and deployed them to **AWS EC2** instances, ensuring consistent execution environments for client reports.

SKILLS

Engineering Domains – Industrial Decarbonisation, Energy Efficiency, Building Services, Mechanical Engineering, Electrical Engineering, Energy Engineering

Data Analysis & Scripting – Python, pandas, NumPy, R, Excel, MATLAB

Engineering Standards – IS 399, EN 16247, ISO 50001, IPMVP

Industrial Systems – HVAC, Steam, Chilled Water, LPHW, Compressed Air, Cooling Towers, Electrical Distribution

Cloud & Containerization – AWS EC2, Docker

Software & Tools – AutoCAD, Revit, Power BI, Tableau, Jira, Confluence

PROJECTS

Industrial Energy Efficiency Analysis Platform (HCL Technologies)

- Developed Python scripts to ingest and analyze energy consumption data from various industrial systems, identifying patterns of inefficiency for a financial services client.
- Built interactive dashboards using **Power BI** to visualize energy usage trends and highlight areas for potential cost savings, presenting findings to operational stakeholders.
- Automated the generation of weekly performance reports using **Python** and SQL queries against PostgreSQL, reducing manual effort by approximately 6 hours per week.

Site Survey Data Aggregation Tool (MSc Project)

- Created a data aggregation tool in Python that processed and standardized data collected from multiple site surveys, including parameters like HVAC operational hours and steam pressure.
- Implemented data validation checks to ensure accuracy and completeness of collected energy system metrics, flagging inconsistencies for review.
- Designed a basic user interface using Flask to allow engineers to input survey data, improving the efficiency of data collection during site visits.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)